

Executive Report

Delivery Delay Prediction Using Machine Learning

Project Overview

This project explores the application of Machine Learning techniques to predict delivery delays within an e-commerce logistics environment using a large-scale Amazon transactional dataset. The objective was to develop a predictive classification workflow capable of identifying whether an order delivery would be delayed based on operational, pricing, inventory, and customer-related variables.

The project was designed to simulate a real-world logistics analytics use case while demonstrating core machine learning concepts including preprocessing, feature engineering, supervised learning, model evaluation, and model comparison.

Dataset Summary

The dataset was sourced from Kaggle and originally contained approximately 1 million rows and 20 columns representing Amazon-style e-commerce transactions.

To improve computational efficiency and maintain project manageability, a working sample of 20,000 rows was selected for model development.

Key Variables Included

- Product category and subcategory
- Product pricing and discounts
- Seller ratings
- Stock availability
- Shipping time
- Customer ratings and review counts
- Payment methods
- Delivery status

The target variable was transformed into a binary classification problem:

- 0 = Not Delayed
 - 1 = Delayed
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Methodology

The project followed a structured Machine Learning workflow:

1. Data Preparation

- Data cleaning and validation
- Duplicate checks
- Feature selection
- Conversion of categorical variables using Label Encoding

2. Feature Engineering

Custom variables were created to enhance predictive analysis:

- `high_rating`
- `fast_shipping`
- `popular_product`

3. Model Development

Two supervised classification models were implemented:

- K-Nearest Neighbors (KNN)
- Random Forest Classifier

4. Model Optimization

Hyperparameter tuning was performed on the KNN model by testing multiple K-values to identify the optimal neighbor count.

5. Model Evaluation

Performance was assessed using:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Feature Importance Analysis

Results & Findings

Model	Accuracy
Initial KNN	64%
Tuned KNN	70.5%
Random Forest	70.8%

Although overall accuracy improved after tuning, evaluation metrics revealed significant class imbalance challenges. The models performed well at identifying non-delayed deliveries but struggled to accurately detect delayed shipments.

This highlighted an important real-world Machine Learning insight:

High overall accuracy does not always indicate strong predictive performance across all classes.

Feature Importance Insights

Feature importance analysis from the Random Forest model identified the following variables as the strongest contributors to delivery outcome predictions:

- Discount levels
- Stock availability
- Product pricing
- Review counts
- Seller ratings

The findings suggest that operational and inventory-related variables may influence delivery outcomes more strongly than customer behavior or product-category-based variables.

Challenges Encountered

Several realistic Machine Learning challenges emerged during the project:

- Class imbalance between delayed and non-delayed deliveries
- Limited predictive signal within available features
- Difficulty identifying delayed deliveries accurately
- Tradeoff between overall accuracy and minority-class recall

These challenges reinforced the importance of critical model evaluation beyond headline accuracy metrics.

Conclusion

This project successfully demonstrated a complete end-to-end Machine Learning workflow within a logistics and e-commerce context. Beyond predictive modeling, the project emphasized analytical thinking, feature interpretation, and practical evaluation of model limitations.

The work provides a strong foundation for future exploration into:

- logistics analytics
- supply chain optimization
- predictive operations modeling
- advanced classification techniques

The project also highlights the growing role of data-driven decision-making within modern logistics and e-commerce operations.

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Tools Used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook, GitHub